

FULL END-TO-END PROJECT PIPELINE (Phase-Wise)

“Multi-Patch Deep Learning + MIL + Explainability + LLM Clinical Reporting for Cancer Detection”

Complete roadmap from start → finish

PHASE 1 — Dataset Acquisition & Setup

1. Choose datasets

Minimum requirement (easiest):

PatchCamelyon (PCam – Kaggle) for patch-level training

Optional (makes project more impressive):

Camelyon16/17 for WSI experience

BreakHis for multiscale classification

2. Download & Organize Data

Recommended folder structure:

```
project/
| — data/
|   | — pcam/
|   | — camelyon16/ (optional)
|   | — breakhis/ (optional)
| — src/
| — models/
| — reports/
| — ui_app/
| — README.md
```

3. Load the .h5 files

train_split.h5

valid_split.h5

test_split.h5

Extract:

X (patch images)

y (labels)

✅ PHASE 2 — Preprocessing Pipeline

1. Color Normalization

Apply Macenko stain normalization to all patches.

2. Tissue Detection / Filtering

Remove blank/background patches using:

Otsu threshold

HSV saturation threshold

3. Patch Augmentations

Use medically appropriate augmentations:

Color jitter

Rotation

Horizontal/vertical flip

Stain augmentation

4. Balancing the dataset

Ensure tumor vs normal patches are balanced.

✅ PHASE 3 — Patch-Level Model Training (Main Backbone)

1. Select model architecture

Start simple → go advanced:

✓ ResNet50 baseline

✓ EfficientNet-B0/B3 (recommended)

✓ Vision Transformer (ViT-Base)

✓ (Optional) Hybrid CNN+Transformer

CNN → feature extractor

Transformer → global reasoning

2. Train patch classifier

Inputs: 96×96 RGB

Output: Tumor vs Normal probability

3. Evaluate

Metrics:

Accuracy

F1-score

AUC-ROC

✓ PHASE 4 — Slide-Level Aggregation (MIL Simulation)

(This is what makes your project “Computational Pathology” instead of simple classification.)

Since PCam is only patches, create bags of patches to mimic WSIs.

1. Group patches into "bags"

Each bag = 20–50 patches

Label = 1 if any patch is tumor

2. Multiple Instance Learning (MIL)

Use an Attention-based MIL model:

Patch features → embedding (from CNN)

Attention layer assigns importance weights

Weighted aggregation → slide-level prediction

3. Slide-level evaluation

Metrics:

Slide-AUC

Slide-F1

Sensitivity @ 95% specificity (clinical standard)

✅ PHASE 5 — Explainability Layer

1. Grad-CAM for CNN

Generate heatmaps:

Tumor activation

Important regions

2. ViT Attention Rollout

Get self-attention maps from transformer heads.

3. Attention MIL Visualization

Show patch-level attention scores:

Red = high importance

Blue = low importance

4. Overlay heatmaps

Overlay on original histopathology patches.

This is mandatory for medical AI (clinically explainable).

✅ PHASE 6 — Optional Generative AI (Virtual Staining)

Add one lightweight generative technique:

CycleGAN → H&E → IHC virtual staining

Stain style transfer

Stain normalization GAN

This directly matches Deepnoid's job description.

✅ PHASE 7 — LLM Integration (VERY Powerful)

To satisfy “LLM + Vision model” requirement.

1. Generate structured JSON outputs

Example:

```
{  
  "tumor_probability": 0.92,  
  "top_regions": ["patch_124", "patch_389"],  
  "explainability": "High activation in lymphocytic clusters.",  
  "heatmap_path": "reports/heatmap_001.png"  
}
```

2. Feed into LLM (e.g., GPT or LLaVA-Med)

Prompt example:

“Generate a pathology-style diagnostic report from the following AI findings...”

3. Output Clinical Report

Example:

“The AI model identifies metastatic carcinoma features with high confidence (92%).
Activation maps highlight malignant epithelial clusters with strong nuclear atypia.
Recommend confirmatory IHC staining and pathology review.”

This is what companies LOVE.

✅ PHASE 8 — Final Deployment & Demo

Create a Streamlit / Gradio UI:

- ✓ Upload patch or slide
- ✓ Show classification result
- ✓ Display heatmap
- ✓ Run LLM to generate report
- ✓ Download PDF report

Optional:

Add "Virtual Stain" button → CycleGAN output

✅ PHASE 9 — Documentation & GitHub Release

Your repo should include:

📌 README with diagrams

Dataflow

Architecture

Explainability pipeline

📌 Jupyter notebooks

Training code

Explainability generation

Virtual staining demo

📌 PDF Research Report

Very professional.

🚀 FINAL PHASE SUMMARY (Copy-friendly)

Phase 1: Dataset download & environment setup

Phase 2: Preprocessing (Macenko normalization, tissue detection)

Phase 3: Patch-level CNN/ViT classifier

Phase 4: Slide-level MIL aggregation (bags + attention)

Phase 5: Explainability (Grad-CAM, ViT attention, MIL heatmaps)

Phase 6: Generative AI (virtual staining) (optional but strong)

Phase 7: LLM clinical report generator

Phase 8: Streamlit/Gradio deployment

Phase 9: GitHub + final report + portfolio polish

🔥 THE 6 TYPES OF IMAGES YOU SHOULD SHOW

(using the PCam dataset you have)

Below is EXACTLY what companies expect in a medical imaging AI portfolio.

✅ 1. Raw Tumor vs Non-Tumor Patches (Simple but required)

Show examples:

- Tumor-positive patch (red text)
- Normal patch (green text)
- 10×10 grid of random samples

These are 96×96 RGB patches.

📌 Files to show:

plots/raw_tumor_examples.png

plots/raw_normal_examples.png

✅ 2. Pixel-Level SEGMENTATION Masks (Yes you can!)

You *do* have:

camelyonpatch_level_2_split_train_mask.h5

This is a **96×96 tumor region mask**.

So you can show:

- Ground truth segmentation mask
- Predicted Grad-CAM / attention heatmap overlay
- Binary mask overlay

📌 Images to show:

✓ Mask alone (black = background, white = tumor)

✓ Patch + mask overlay

✓ Patch + Grad-CAM + mask triple overlay

BIG impact. Hiring managers love this.

✅ 3. Grad-CAM Heatmaps on Patches (Explainability)

This is the #1 most important visual for AI pathology.

For tumor patches:

- red = high tumor activation
- blue = low activation

For normal patches:

- model activation should be near zero

📌 **Images to show:**

plots/gradcam_tumor_example.png
plots/gradcam_normal_example.png

✅ **4. Attention MIL Heatmaps (Slide-Level Representation)**

Since you don't have real WSIs, you simulate WSIs using patch groups.

Show:

- A grid of patches from a single WSI
- A heatmap overlay of MIL attention weights

Example:

-  high tumor probability
-  medium
-  low

📌 **Images to show:**

plots/mil_slide_attention.png

Recruiters will think you worked with WSI-level modeling.

✅ **5. Slide-Level Bag Visualization (MIL Step)**

Simulate a slide by arranging patches in coordinates based on metadata.

You have:

train_metadata.csv

→ This contains x, y coordinates for each patch!

So you can reconstruct a **mini-slide layout** like:

```
[patch][patch][patch][patch]
[patch][patch][patch][patch]
[patch][patch][patch][patch]
```

Then overlay:

- tumor patches in red
- normal patches in blue

📌 **Images to show:**

plots/slide_bag_visualization.png

✅ **6. Final Diagnostic Report Screenshots (LLM)**

Not real images but included in UI.

- Predicted tumor probability
- Suggested region of interest
- LLM-generated pathology report

✅ PHASE 1 — Dataset Exploration Visuals

1. Tumor vs Normal Sample Patches

- raw_tumor_examples.png
- raw_normal_examples.png
- raw_tumor_normal_grid.png (a 5×10 grid)

Why: Shows dataset understanding and pathology domain awareness.

✅ PHASE 2 — Preprocessing Visuals

2. Color Normalization Before/After (Macenko)

- macenko_before_after.png

Left = raw patch

Right = normalized patch

3. Tissue Detection Mask

- tissue_detection_mask.png

Show:

- original patch
- binary tissue mask

4. Data Augmentation Examples

- augmentation_examples.png

Show transformations:

- color jitter
 - rotation
 - stain augmentation
-

✅ PHASE 3 — Patch-Level CNN/ViT Model Visuals

5. Grad-CAM Heatmaps

For tumor patches:

- gradcam_tumor_1.png
- gradcam_tumor_2.png

For normal patches (should show almost no activation):

- gradcam_normal_1.png
- gradcam_normal_2.png

6. Grad-CAM Overlay Comparison

- gradcam_overlay_comparison.png

Layout:

- RAW patch
- Grad-CAM (heatmap)
- Overlay

✅ PHASE 4 — MIL (Multiple Instance Learning) Visuals

This is the MOST IMPORTANT part.

7. Bag (Slide) Reconstruction via Metadata Coordinates

You can plot an approximate “slide” using patch coordinates:

- wsi_reconstructed_bag.png

Visualization:

- ☐ normal patch
- ☒ tumor patch

8. MIL Attention Heatmap Over Slide Grid

- mil_attention_wsi_heatmap.png

Shows importance/weights for each patch in the bag.

9. Patch-Level Attention vs Prediction Plot

- mil_patch_attention_ranked.png

Plot top-20 most important patches with scores.

✅ PHASE 5 — Pixel-Level Mask Visuals (SEGMENTATION)

Since you have:

camelyonpatch_level_2_split_train_mask.h5

You can show:

10. Ground Truth Mask

- mask_ground_truth.png

11. Mask Overlaid on Patch

- patch_mask_overlay.png

12. Grad-CAM / Mask Comparison

- mask_vs_gradcam.png

Left = ground-truth tumor region

Right = Grad-CAM heatmap overlay

★ This is extremely powerful — proves interpretability quality.

13. IoU / Dice Score Visualization

- mask_iou_distribution.png

Plot model segmentation alignment quality.

✅ PHASE 6 — (Optional) Virtual Staining / Style Transfer

14. Stain Transfer Example

(If using CycleGAN)

- virtual_stain_he_to_ihc.png

Before = H&E

After = Virtual IHC

15. Stain Normalization GAN Demo

- stain_normalization_example.png
-

✅ PHASE 7 — LLM Report Visualization

16. Clinical Report UI Screenshot

- llm_report_ui.png

17. JSON Data → LLM Generated Report

- llm_generated_pathology_report.png
-

✅ PHASE 8 — Final Inference UI (Streamlit/Gradio)

18. Complete Inference Screen

- ui_full_pipeline_screenshot.png

Includes:

✓ Input patch

✓ Tumor probability

✓ Grad-CAM heatmap

✓ MIL attention (if slide-level)

✓ Clinical report

19. Slide-Level Heatmap Panel

- ui_slide_attention_panel.png